Cambridge
O Level

**Cambridge Assessment International Education**
Cambridge Ordinary Level

**PHYSICS** **5054/42**

Paper 4 Alternative to Practical **October/November 2017**

MARK SCHEME

Maximum Mark: 30

**Published**

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| Question | Answer | Marks |
|---|---|---|
| 1(a)(i) | 4.7 (cm) correct answer only | **B1** |
| 1(a)(ii) | 47(.0) cm unit required | **B1** |
| 1(b)(i) | 1.14, 1.30 2 d.p. required in both | **B1** |
| 1(b)(ii) | to reduce the effect of errors in starting / stopping the stopwatch / to reduce the effect of reaction time / to calculate an average / to reduce the percentage error | **B1** |
| 1(c)(i) | axes labelled quantity **and** unit axes correct way round | **B1** |
| | scales linear, not awkward, start from (0,0) | **B1** |
| | points plotted accurately | **B1** |
| | smooth best fit straight line drawn | **B1** |
| 1(c)(ii) | candidate's value to ±½ small square from a correct extrapolation | **B1** |
| 1(c)(iii) | values indicated on graph or triangle drawn and ⩾ half the range of the plotted points ($\Delta h \geqslant 12.5$) | **B1** |
| | $m$ = (−) 0.04 ± 0.01 | **B1** |
| 1(d) | candidate's $c$ / $m$ value correctly calculated | **B1** |
| 1(e) | YES (must be stated) **and** values very close / close enough / within the limits of experimental error / < 10% **OR**<br>NO (must be stated) **and** values not close / not close enough / outside the limits of experimental error / > 10% | **B1** |

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| Question | Answer | Marks |
|---|---|---|
| 2(a)(i) | 88 (°C) c.a.o. | **B1** |
| 2(a)(ii) | to allow thermometer to read the maximum temperature of the hot water / to give the thermometer time to respond / to wait until the temperature on thermometer stops rising / so that temperature of thermometer equals temperature of the water / to allow thermometer to reach thermal equilibrium | **B1** |
| 2(b)(i) | s, °C, °C | **B1** |
| 2(b)(iii) | significant change **and** temperature drops are not close / similar / there is a 3 °C difference **or**<br>not significant change **and** temperature drops are close / similar / there is only a 3 °C difference | **B1** |
| 2(c)(i) | use a lid / cover the beaker | **B1** |
| 2(c)(ii) | lag the bottom of the beaker / use thicker lagging | **B1** |
| 2(d) | any one from: same volume of (hot) water / same (size) beaker / room temperature / same time (of cooling) | **B1** |

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| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | normal drawn at point Q **and** extended to cross CD with point R correctly labelled | **B1** |
| 3(a)(ii) | $\alpha = 30 \pm 1°$<br>unit required | **B1** |
| 3(b)(i) | straight line through crosses, crossing CD at S and QR at T, S and T both labelled<br>**and** $x = 5.3 \pm 0.1$ (cm) | **B1** |
| 3(b)(ii) | $y = 3.5 \pm 0.1$ (cm) | **B1** |
| 3(c) | candidate's $x$ / $y$ calculated correctly | **B1** |

| Question | Answer | Marks |
|---|---|---|
| 4(a) | wire, ammeter, power supply in series and voltmeter in parallel with the wire (or the cell, if the only resistance in the circuit is provided by the wire) | **B1** |
| | symbols for cell / battery / power supply, ammeter and voltmeter correct | **B1** |
| 4(b) | measure $V$ and $I$ (and calculate $R$) | **B1** |
| | repeat for different lengths | **B1** |
| 4(c) | plot a graph of $R$ against $l$ (can be credited from b) / (use table / readings to) compare resistance values as length changes | **B1** |

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